Object Detection

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This document demonstrates **Ultralytics** Object Detection with Python, showing its performance on images, videos, and real-time detection.

1、Enable Optimal Board Performance

1.1、Enable Jetson Clocks

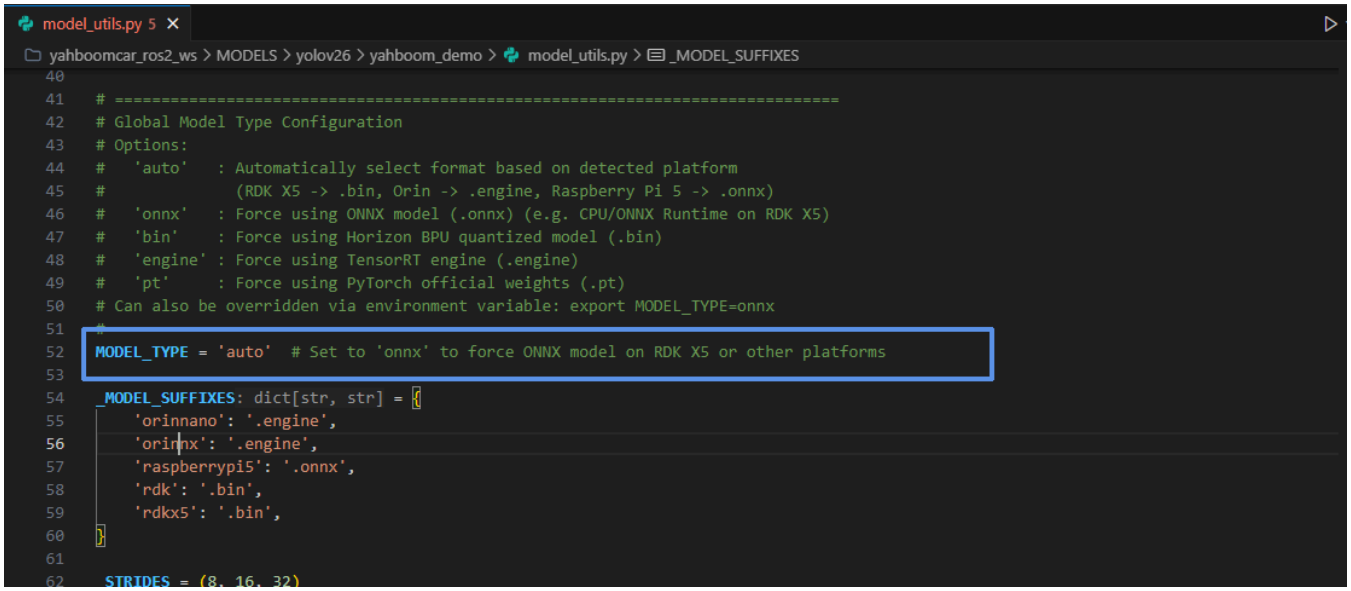
Enabling Jetson Clocks ensures that all CPU and GPU cores run at maximum frequency: **(Not required on RDK and Raspberry Pi boards)**

```
sudo jetson_clocks
```

RDK boards can use the officially pre-quantized models for BPU inference. If required, you can also choose to use ONNX for inference — just modify the following location:

```
~\yahboomcar_ros2_ws\MODELS\yolov26\yahboom_demo\model_utils.py
```

Change it to ONNX to use the ONNX model.



2、Object Prediction: Images

Use yolo26n.pt to run prediction on the images bundled with the ultralytics project.

Navigate to the code folder:

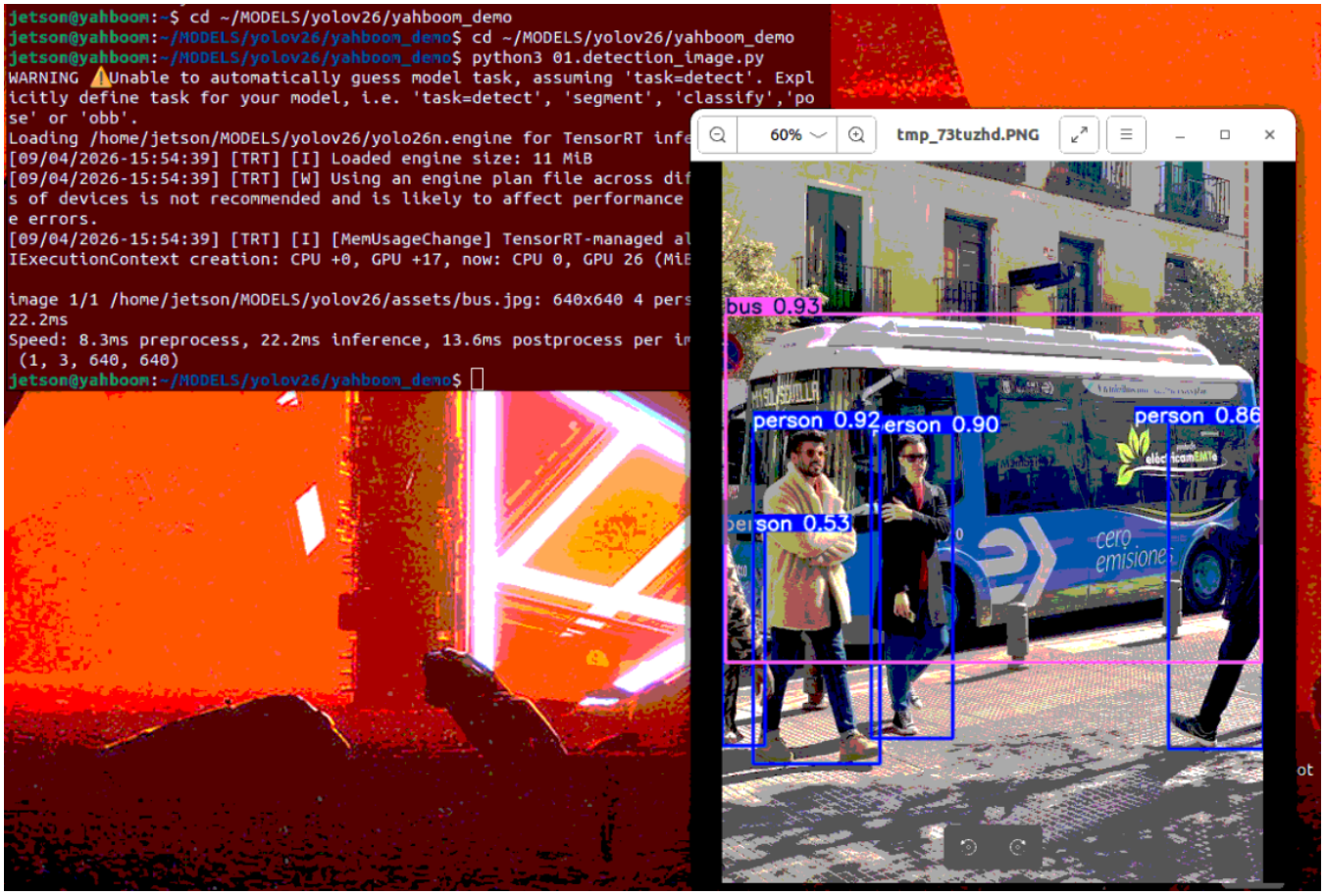
```
cd ~/MODELS/yolov26/yahboom_demo
```

Run the code:

```
python3 01.detection_image.py
```

Preview

Location of the images output by yolo: ~/MODELS/yolov26/assets/output/



3、Object Prediction: Video

Use yolo26n.pt to run prediction on a video in the ultralytics project (a video not bundled with ultralytics).

Navigate to the code folder:

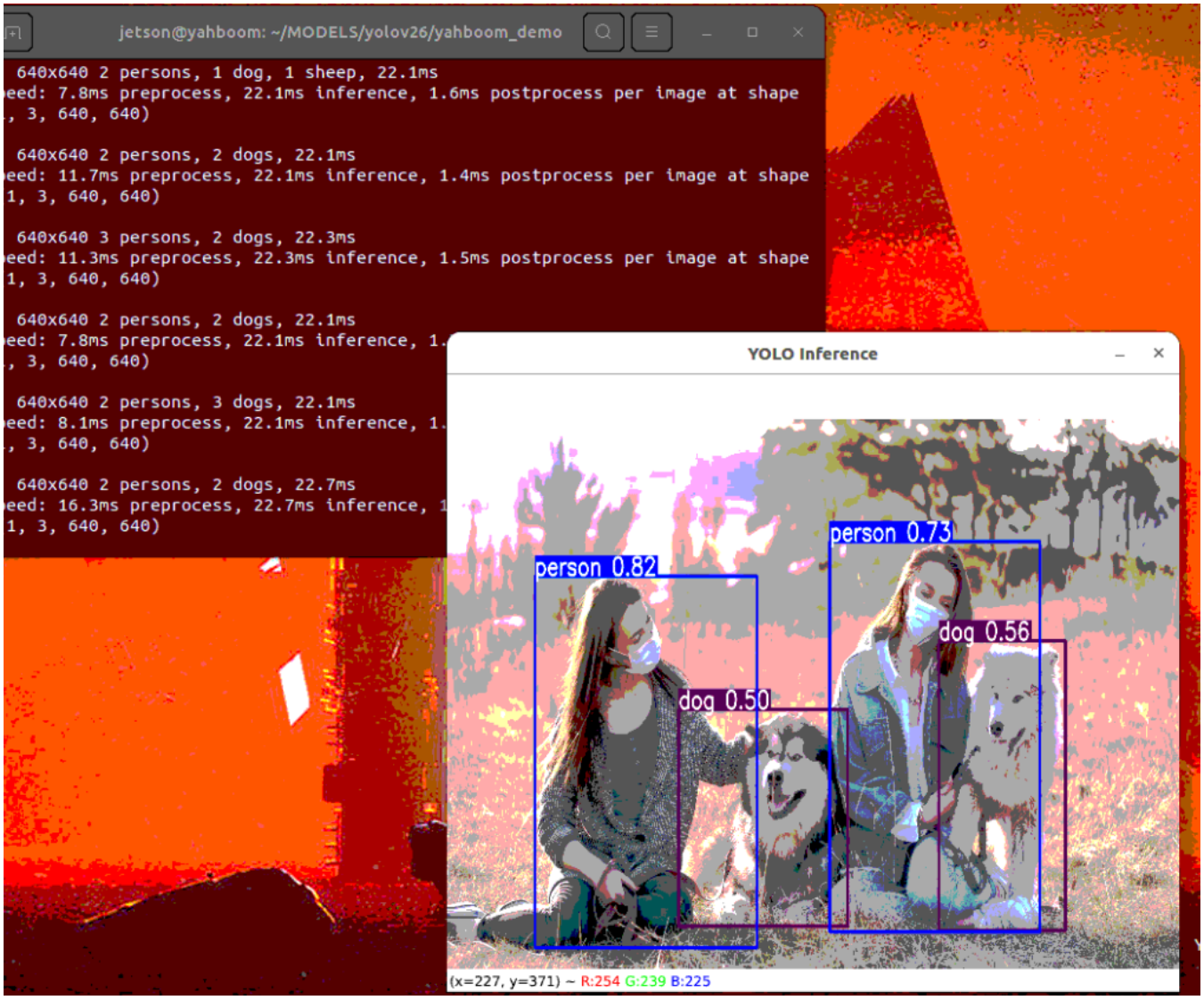
```
cd ~/MODELS/yolov26/yahboom_demo
```

Run the code:

```
python3 01.detection_video.py
```

Preview

Location of the videos output by yolo: ~/MODELS/yolov26/videos/output/



4、Object Prediction: Real-Time Detection

4.1、Camera

Use yolo26n.pt to run prediction on the USB camera feed.

Start the camera:

```
#USB camera
ros2 launch usb_cam camera.launch.py
#Nuwa camera
ros2 launch ascamera hp60c.launch.py
```

Navigate to the code folder:

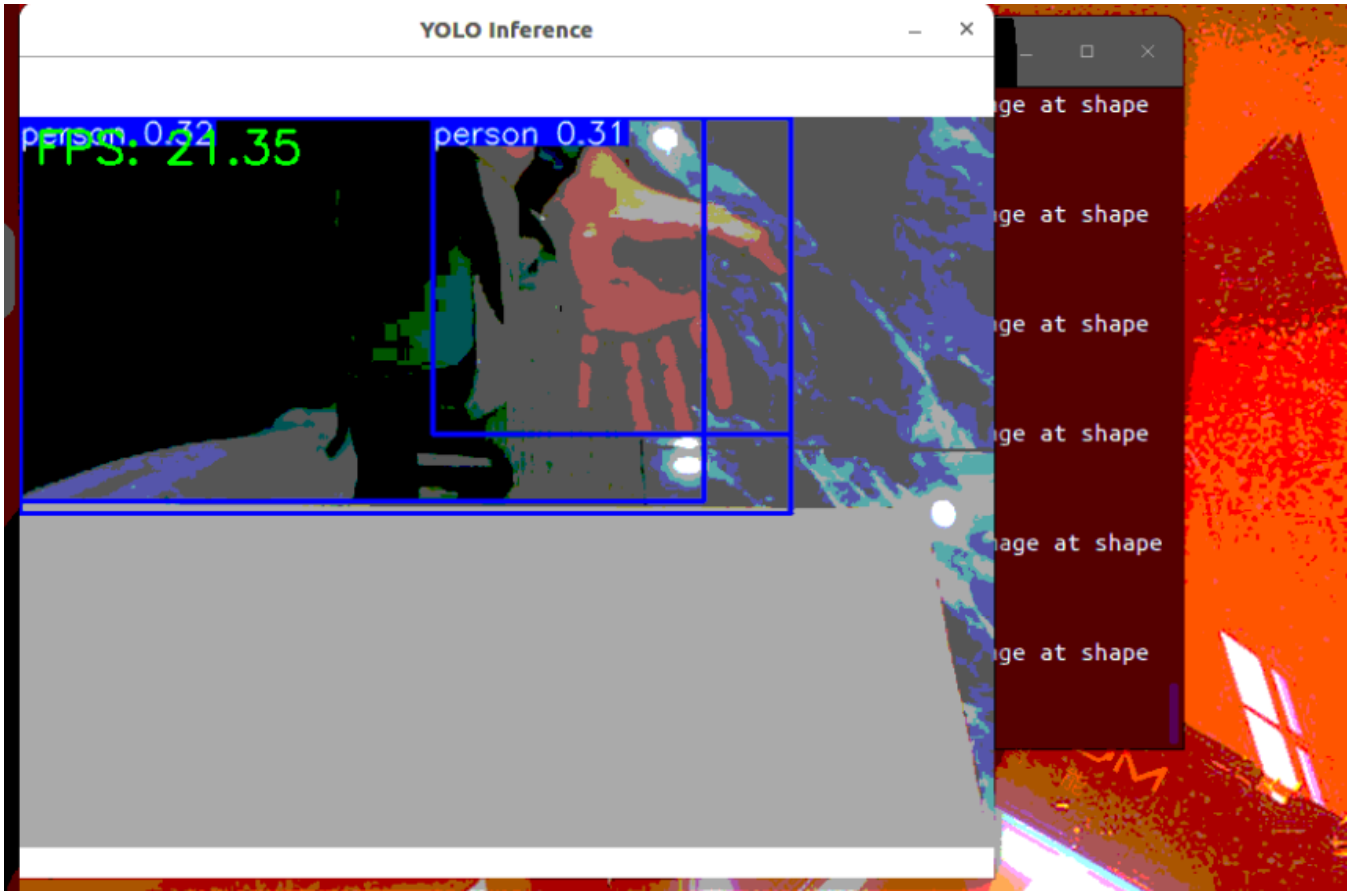
```
cd ~/MODELS/yolov26/yahboom_demo
```

Run the code: click the preview window and press the q key to terminate the program!

```
python3 01.detection_camera_usb.py
```

Preview

Location of the videos output by yolo: ~/MODELS/output/ **(On RDK and Raspberry Pi boards, refer to the actual path printed in the terminal)**



References

https://docs.ultralytics.com/tasks/detect/